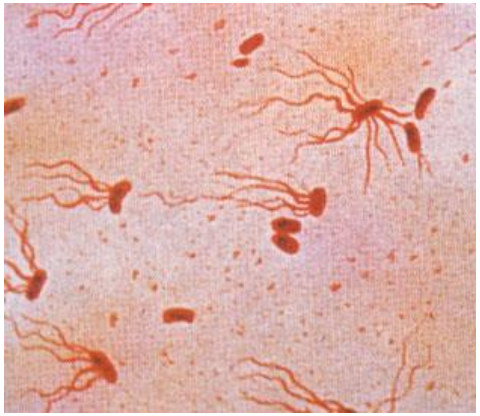
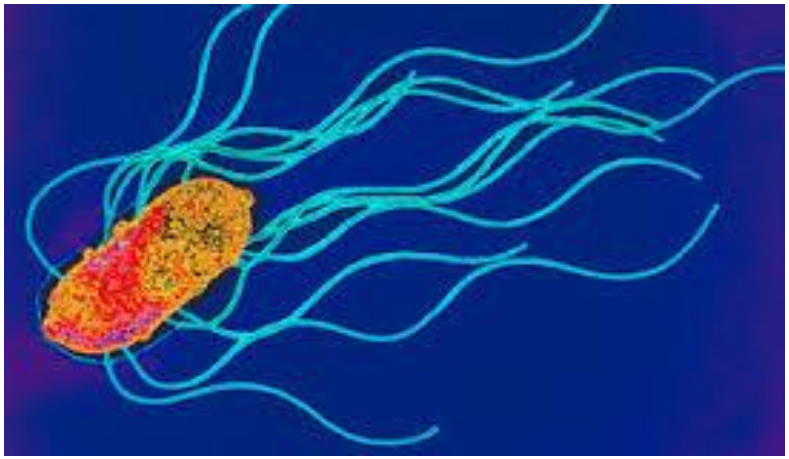
Primary intestinal pathogens of the order Enterobacterales

- *Salmonella* spp.
- *Shigella* spp.
- *Yersinia* spp.
- *Plesiomonas shigelloides*



Salmonella spp.

(Family Enterobacteriaceae)

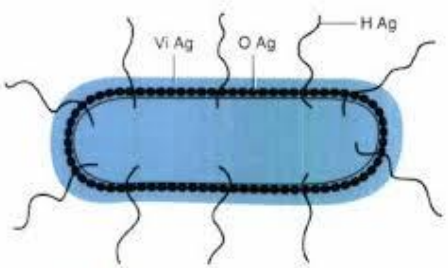


Classification

- There are only 2 species:
  - *S. enterica*; the type species of the genus
    - subsp. *enterica* (subsp. I)
    - subsp. *salamae* (subsp. II)
    - subsp. *arisonae* (subsp. IIIa)
    - subsp. *diarizonae* (subsp. IIIb)
    - subsp. *houtenae* (subsp. IV)
    - subsp. *indica* (subsp. VI)
  - *S. bongori*; rarely isolated species from cold-blooded animals and the environment

General characteristics

- Gram-negative, facultative anaerobic rods
- Non-lactose fermenter; oxidase negative
- Motile by peritrichous flagella
- Most produce H_2S, except *S.* Paratyphi
- Possess (heat-stable) O and (heat-labile) H antigens that used in serologic grouping; a few strains may posses (heat-labile) K or Vi antigens



General characteristics

- Most clinical significant isolates belong to *S. enterica* subsp. *enterica*
  - >2,500 serotypes การจัดระบุตัวเชื้อ โดยใช้ Antigen - Antibody reaction
    - *S.* Enteritidis, *S.* Typhi, *S.* Paratyphi, *S.* Cholerasuis, *S.* Typhimurium
- *Salmonella* is classified into serogroups based on differences in their surface O antigens.
  - >60 serogroups
    - S. Enteritidis (group B or E), S. Typhi (group D), S. Paratyphi A (group A) , S. Cholerasuis (group C), S. Typhimurium (group B)

Differences between serogroup and serotype

Characteristic	Serogroup	Serotype
Definition	A classification based on shared **O antigens** (lipopolysaccharide)	A specific strain within a serogroup based on both **O** and **H antigens** (lipopolysaccharide and flagellar proteins)
Scope	Broader classification grouping strains with similar **O antigens**	A more specific classification of a strain with distinct **O** and **H antigens**
Level of Specificity	Less specific (groups similar strains)	More specific (identifies individual strains within a group)
Example	**Serogroup D** (includes **Typhi**, **Paratyphi D**)	**Serotype Typhi** (from **serogroup D**)
Number of Strains	Can include multiple **serotypes**	Represents one specific **strain** or variant

Virulence factors

- ① Frimbriae: ยึดติดในชั้น mucosa ใน small intestine
  - Adherence to the mucosa of small intestine
- ② Enterotoxin: เกิดอาการท้องเสีย เพราะดูดน้ำ
  - Alters permeability of intestinal wall
  - Water leaks into intestinal tract causing diarrhea
- ③ Endotoxin: ถูกปล่อยมา when cell แตก → ไข้
  - Released by bacterial cell lysis
  - Causes fever and inflammation
- ④ Capsular (K) antigen ป้องกัน phagocytosis
  - Prevent phagocytosis

Clinical Manifestations

- Infective dose: 10^5-10^{10} bacteria
- In humans, salmonellosis may occur in several forms:
  - 1 An acute gastroenteritis or food poisoning
    - The most common presentation of nontyphoidal *Salmonella* infection
    - ! *S.* Enteritidis is the most common pathogen.
    - **Nausea**, **vomiting**, fever and chills, accompanied by **nonbloody diarrhea** and abdominal cramps
    - Most cases are self-limiting.
    - Ingestion of organisms through contaminated food
    - Incubation period: ~6-72 h

Clinical Manifestations

ไข้รากสาดน้อย

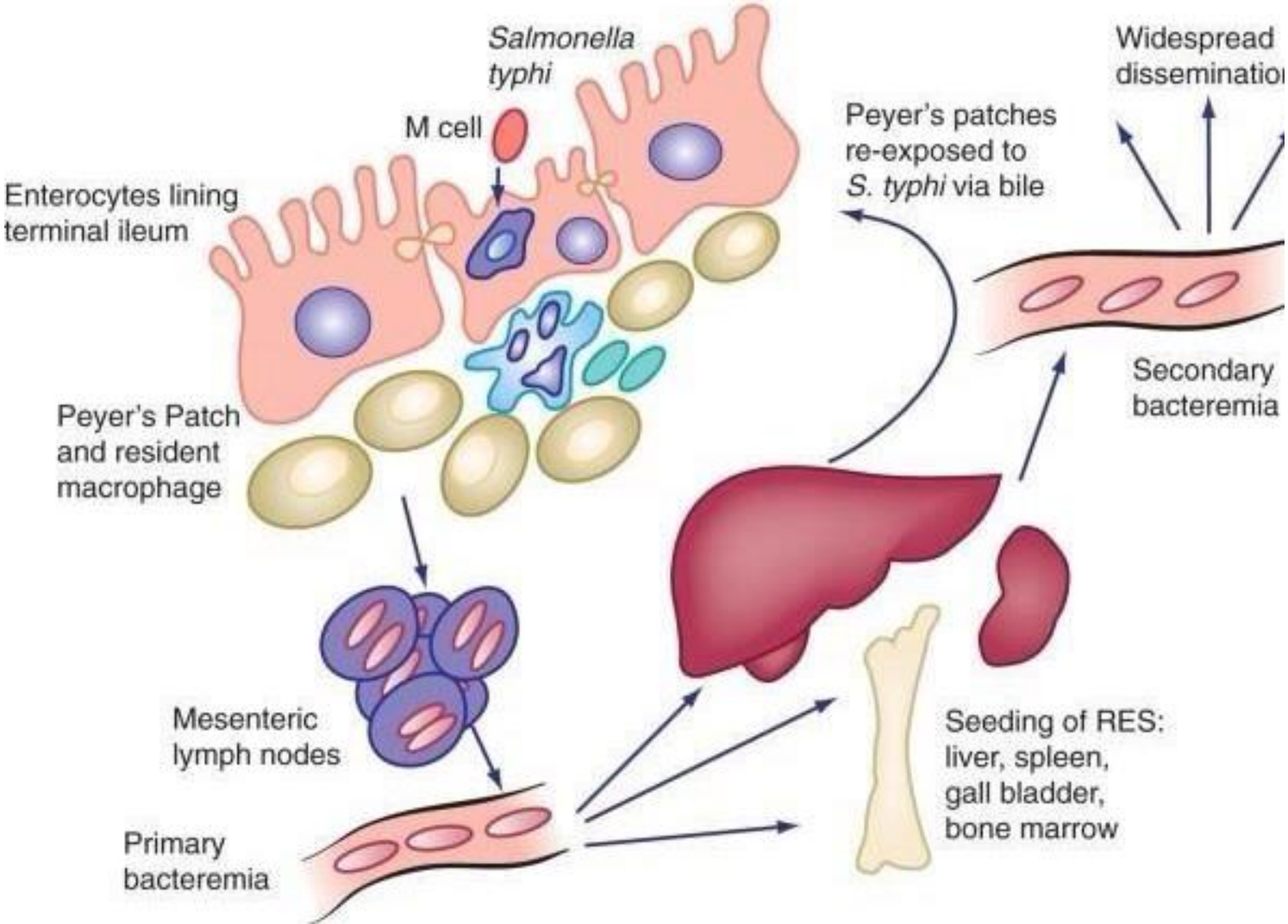
– Enteric fever

- Prolonged fever, malaise, anorexia, lethargy, myalgia, headache, bacteremia, involvement of reticuloendothelial system (liver, spleen, intestines and mesentery) and dissemination to multiple organs
- **Typhoid fever** caused by *S.* Typhi and **paratyphoid fever** caused by other *Salmonella* (e.g. *S.* Paratyphi A, B and C).
- Humans are the only source of infection.
- Ingestion of food contaminated with organisms originated from infected individual or carriers

Clinical Manifestations

- Typhoid fever develops approximately 7-14 days after ingestion of the organisms.
- During the 2nd and 3rd weeks, patient experiences sustained fever with prolonged bacteremia
- Rose spots appear during the 2nd week of fever.
- Diarrhea is intermittent or absent.
- The organisms invade the Payer's patches of the bowel and gallbladder. They also reach the intestinal tract via the biliary tract and reinfect. The gallbladder become the foci of long-term carriage of organisms, occasionally shedding the organisms in the feces.

Pathogenesis of typhoid fever



Pathogenesis of typhoid fever

- After ingestion, organisms attach to the mucosa of **small intestine** and invade into the **microfold (M) cells**, as well as into enterocytes.

* Microbe อยู่ใน Endocytic vacuoles can ข้ามผ่าน cytoplasm เข้าสู่ เลือด และ น้ำเหลือง *

- The organisms remain in endocytic vacuoles. They can also be transported across cytoplasm and released into the blood or lymphatic circulation.

แพร่ ไปที่ ตับ ม้าม ไขกระดูก

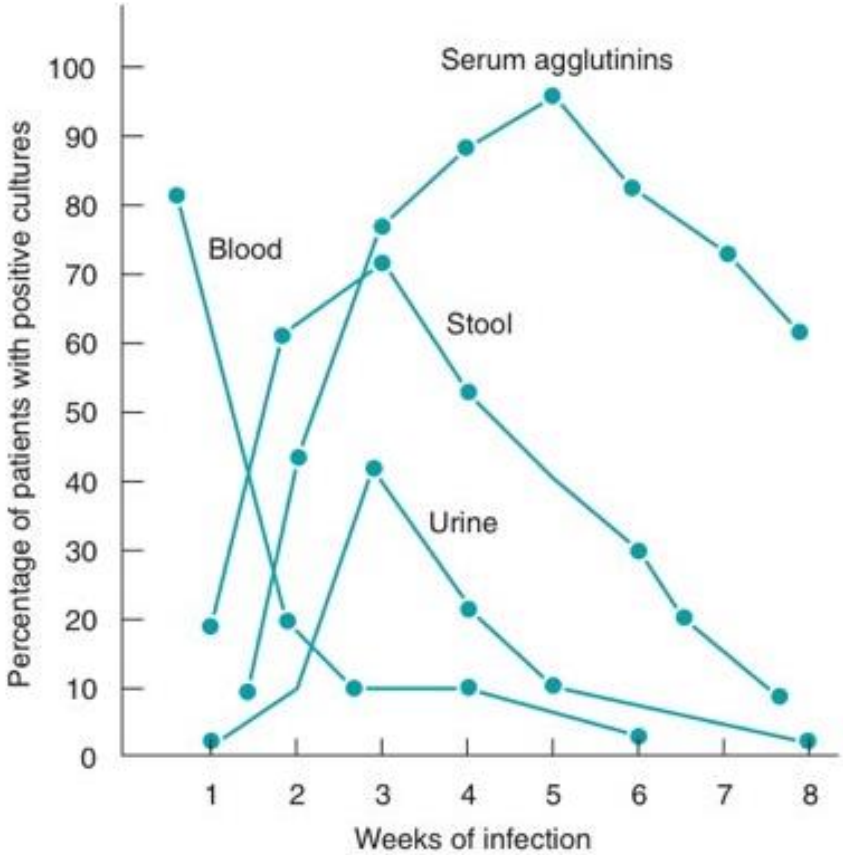
- They are further spread to the liver, spleen and bone marrow, where they are immediately engulfed by mononuclear phagocytes.

เพิ่มจำนวน ภายใน cell ปล่อยเข้าสู่กระแสเลือด

- The organisms multiply intracellularly; later they are released into the blood stream for the second time

- Bacteria invade gall bladder, biliary system and lymphatic tissue of the bowel and multiply in high number.

Culture and serologic diagnosis of typhoid fever



Clinical Manifestations

3

- Nontyphoidal septicemia
  - Prolonged fever and intermittent bacteremia
  - *S.* Typhimurium and *S.* Cholerasuis are the most common.
  - Clinical presentation is like other Gram-negative bacteremia.

4

- Carrier state (Asymptomatic colonization)
  - Individuals who recover from infection may harbor the organisms in the **gallbladder** (site of chronic carriage) and excrete organisms in their feces.

Epidemiology

จากสัตว์ปีก กินไก่ กินไข่ดิบ หรือ ใน สัตว์เลี้ยง ผลิตภัณฑ์นม

- ! Salmonellosis is a zoonosis. Wild and domestic poultry serve of as an important reservoir.
- Contaminated food products (**poultry**, **eggs**, and **daily products**) are the most common sources of infection.
- Direct fecal-oral spread in children
- ! *S.* Typhi and *S.* Paratyphi are **strict human pathogens** ! (no reservoirs); these infections are passed **person to person**; asymptomatic long-term colonization occurs commonly ! (pathogen can survive in gallbladder and establish chronic carriage) !

Epidemiology

ผู้ป่วยที่มีความเสี่ยง

① ผู้ที่ติดเชื้อไข่ไก่ไม่สุก

- Individuals at risk for infection include those who eat improperly cooked poultry or eggs, patients with reduced gastric acid levels, and immunocompromised patients. ② ผู้ป่วยที่มี LV. กรดในกระเพาะ ลดลง และ ภูมิคุ้มกันบกพร่อง
- The risk of disease is highest in children living in a developing country.
- Infections occur worldwide, particularly in the warm months of the year.

Laboratory diagnosis

- Blood, bone marrow and stool culture are the most reliable diagnostic methods for enteric fever.
- Isolation from stool specimens or rectal swab requires use of selective media (e.g. MacConkey, XLD and SS media).
- Enrichment media for salmonellae: selenite F (SF), tetrathionate (TT), Rappaport-Vassiliadis (RV) broth
- Biochemical tests are used for identification and O serogroup antisera are used for comfirmation.

MacConkey agar

XLD agar

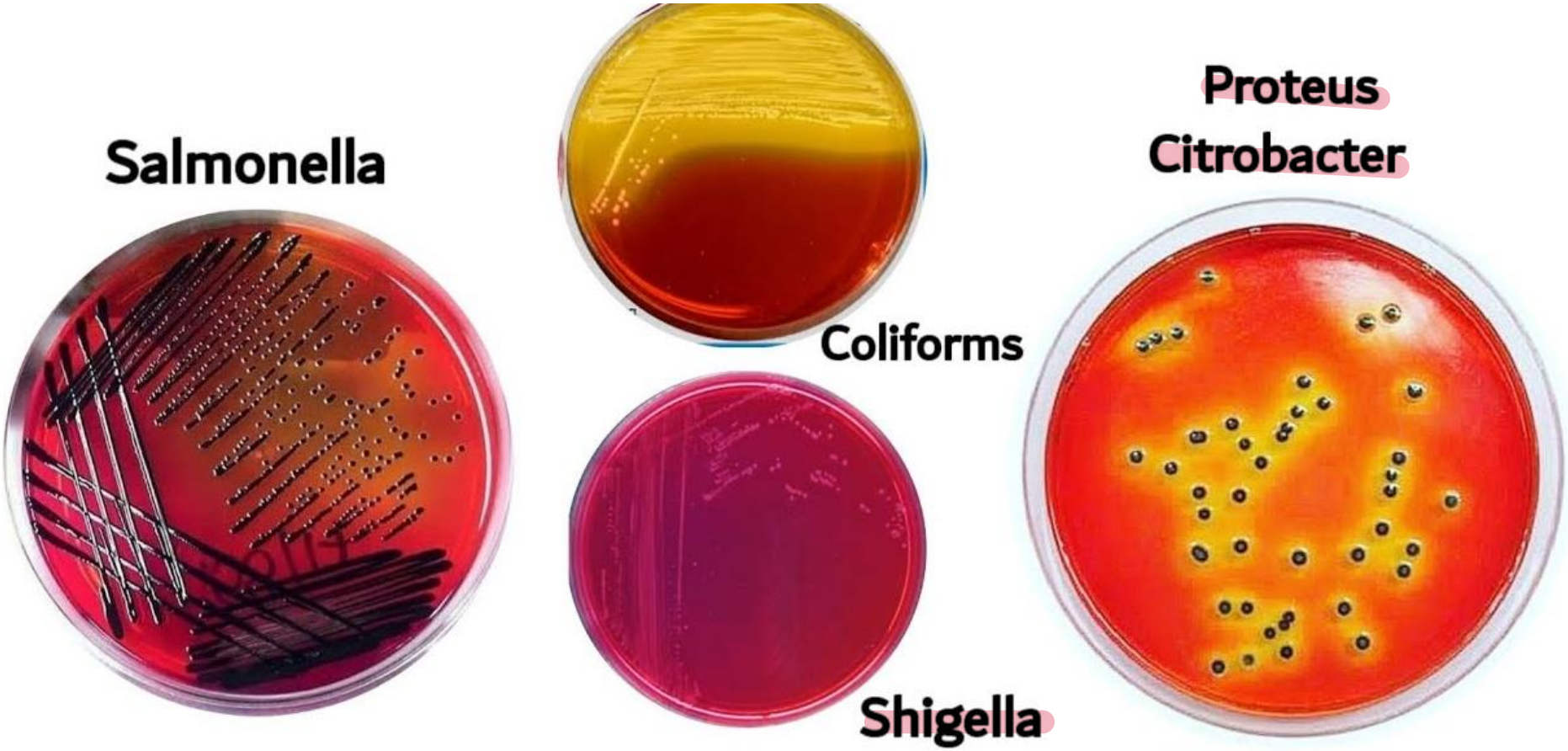
Xylose-lysine deoxycholate (XLD) agar

- Selective and differential agar used for detection of *Salmonella* and *Shigella*
- Sodium deoxycholate inhibits the growth of Gram-positive bacteria.
- Those that grow typically ferment lactose, sucrose or xylose producing yellow colonies (phenol red as indicator).
- *Salmonella* ferments xylose but also decarboxylates lysine, thus the colonies appear red. *Salmonella* also produce H_2S from sodium thiosulfate, colonies will turn black in the presence of ferric ammonium citrate.

  จับตัวเป็น Colony ดำ

  $Na_2S_2O_3$
- *Shigella* does not ferment these sugars, so the colonies appear red.

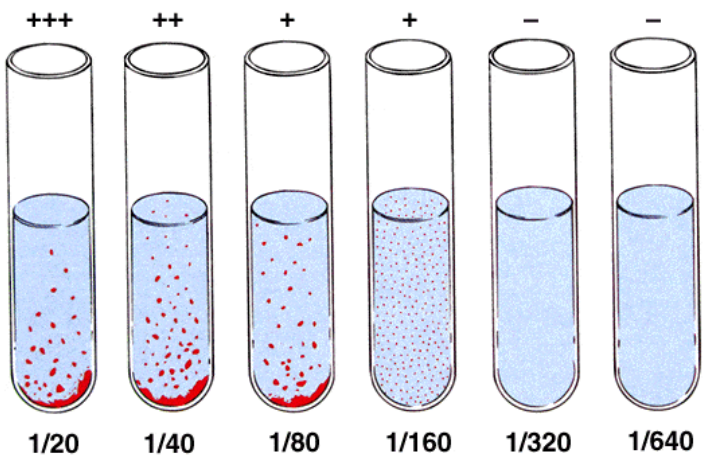
XLD agar



ให้สีแดง

Laboratory diagnosis

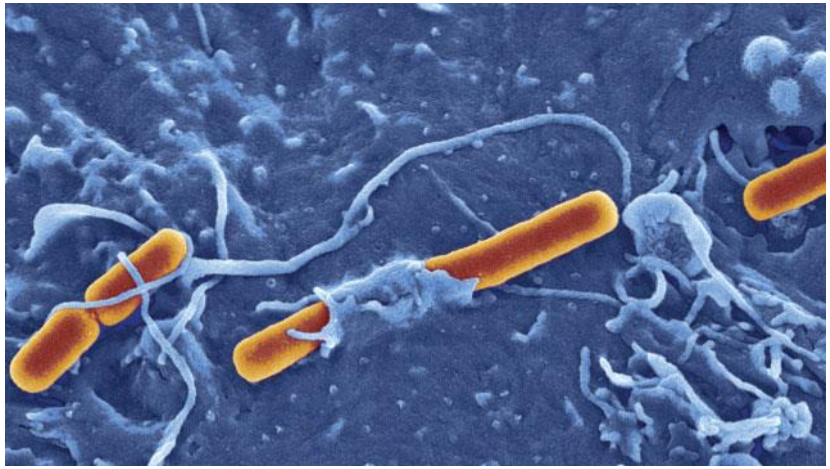
- The Widal test detects agglutinating antibodies against the O and H antigens of *S.* Typhi and S. Paratyphi.
- This test is cheaper and requires minimal equipment. However, it is considered insensitive and nonspecific.
- The Widal agglutination test has been largely replaced by enzyme-linked immunosorbent assay (ELISA) and immunoblotting.



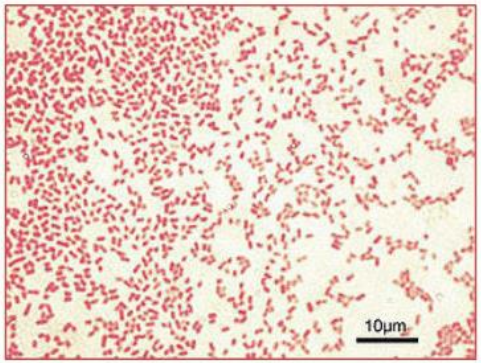
Treatment, prevention and control

- Antibiotic treatment not recommended for enteritis.
- Antimicrobial therapy is clearly indicated in typhoid fever.
- Ciprofloxacin, ceftriaxone and cefixime are preferred as first-line agents.
- Most infections can be controlled by preparation of poultry and eggs and avoidance of contamination of other foods with uncooked poultry products.
- Carriers should be identified and treated.
- Vaccination against *S.* Typhi can reduce the risk of disease for travelers into endemic areas.

Shigella spp.
(Family Enterobacteriaceae)



General characteristics



- Gram-negative, facultative anaerobic rods
- NLF
- Non-lactose fermenter; oxidase negative
- oxidase ⊖
- Do not produce H_2S
- ! Non-motile !
- urease ⊖ LD ⊖ gas from glu ⊖
- They are negative for urease, LD and gas from glucose
- All possess O and certain strains can possess K antigens
- They are fragile organisms, that are susceptible to various physical or chemicals agents and acid pH.

แปรสะบาง และ ไวต่อ physical หรือ chemical

Classification

- According to O antigens, there are 4 species:
  - *S. dysenteriae* (serogroup A) → produces a severe illness
    mannitol (-)
  - *S. flexneri* (serogroup B) → most common in developing countries
    ใช้ sero แยก
  - *S. boydii* (serogroup C) → uncommon
  - *S. sonnei* (serogroup D) → most infections in developed countries
    ONPG (+)
    - ~50 serotypes
    - Molecular analysis determined that 4 species are so closely related to *E. coli.*

Virulence factors

จับกับตัว Host cell

- Genes for adherence, invasion, intracellular replication and cell-to-cell spread
- Endotoxin

  ทำให้ large intestine เกิดการระคายเคือง

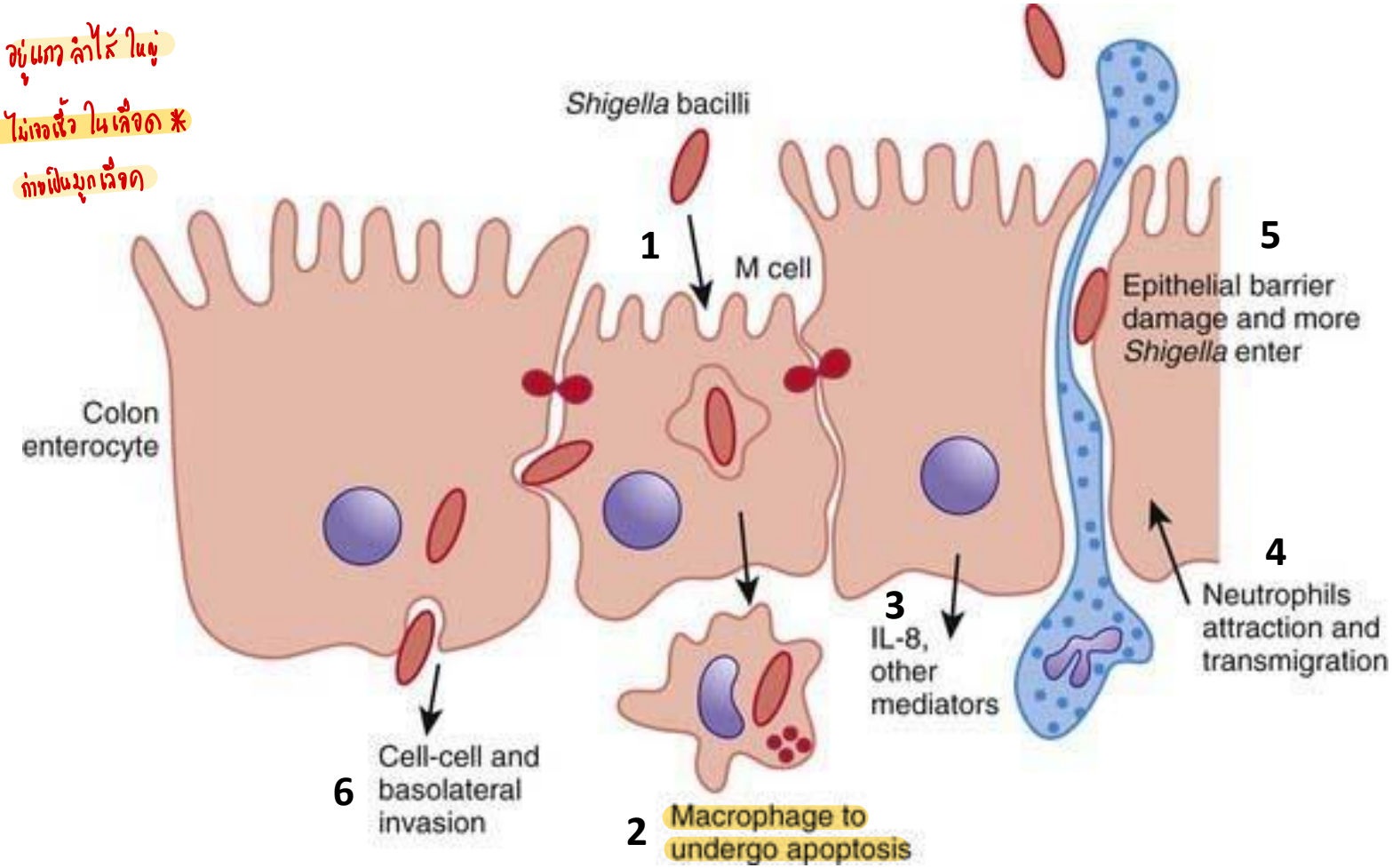
  – Contributes to the irritation of the bowel wall
- Exotoxin (Shiga toxin)
  – Is a heat-labile exotoxin produced by *S. dysenteriae*; disrupt protein synthesis and produces endothelial damage
  – Affects the gut (enterotoxin), CNS (neurotoxin) and kidney
  – Causes hemorrhagic colitis (HC) and hemolytic uremic syndrome (HUS)

Pathogenesis of *Shigella* spp.

อยู่แถวลำไส้ ใหญ่

* ไม่เจอเชื้อ ในเลือด *

ก่อเป็นมุกเลือด



Pathogenesis

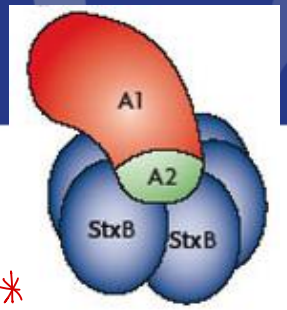
- Infective dose: 10-100 bacteria
- Shigellae cause disease by invading and replicating in cell lining the **colon**.
- They first attach and invade M cells located in Peyer's patches.

หลั่ง Protein ใน Target cell → กิน

- They secrete the proteins on the target cells, leading to engulfment of bacteria.

ช่อง phogocytic vacuoles และเพิ่มจำนวนใน cytoplasm

- They lyse the phogocytic vacuoles and replicate in host cell cytoplasm.
- With the rearrangement of actin filaments, the bacteria are propelled through the cytoplasm to adjacent cells. They are protected from immune-mediated clearance.



Pathogenesis

* รอดจาก phogocytosis โดย เหนี่ยวนำ apoptosis → ปล่อย cytokines ดึงดูด PMN เข้าไปใน infected tissue *

- They survive phagocytosis by inducing **apoptosis** and lead to the release of cytokines resulting in the attraction of PMN into the infected tissues.

- **Shiga toxin**: commonly found in *Shigella* dysenteriae serotype 1 that can cause serious human diseases

  B จับ host cell → A เข้า cell

  – B subunits bind to the host cell glycolipid (Gb3) and facilitate transfer of the A subunit into the cell.

  – A subunit cleaves the 28S rRNA and disrupts protein synthesis.

  A → ✗ protein synthesis

  ทำลาย epithelium ของลำไส้ และ glomerulay → make HUS

  – Damage to intestinal epithelium and glomerular endothelial cells, resulting in renal failure (HUS)

Clinical Manifestations

โรคบิดไม่มีตัว

- Most common form of disease is gastroenteritis (shigellosis); **fever**, an initial watery diarrhea progressing within 1 to 2 days, **abdominal cramps** and **tenesmus** with **bloody stool** (blood and mucus).

  ปวดเบ่ง stool มีมูกเลือด

  - The first sign of infection is mediated by enterotoxin.

    ทำให้เกิด

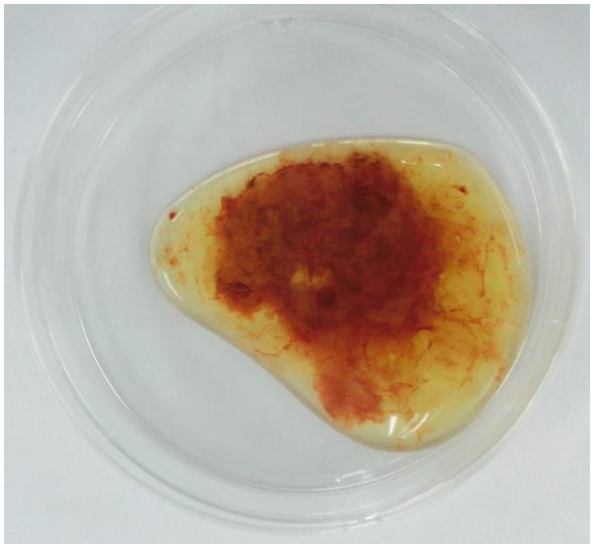
  - The bloody stool results from invasion of the colonic mucosa by the bacteria.

- Severe form of disease is caused by *S. dysenteriae* (bacillary dysentery)

  พาหะไม่แสดงอาการ มีจำนวนน้อย

- Asymptomatic carriage develops in a small number of patients.

Clinical Manifestations

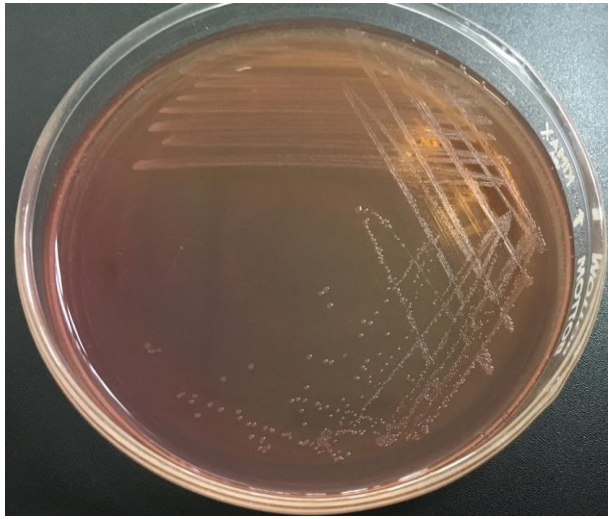


Epidemiology

- They are **strict human pathogen** and **humans** are the reservoir.
- Disease **spread person to person** by fecal-oral route or indirectly through contact with water or food
- Patients at highest risk are young children.
- Relatively few organisms can produce disease (highly infectious/low infective dose).
- Disease occurs worldwide with no seasonal incidence.
- Shigellosis spread rapidly in communities where sanitary standards and the level of personal hygiene are low.

Laboratory diagnosis

- Isolation from stool specimens or rectal swab requires use of selective media (e.g. MacConkey, XLD and SS media).
- Enrichment media for Shigellae: Gram negative (GN), Shigella (SB), selenite F (SF) broth
- Biochemical tests are used for identification and O serogroup antisera are used for comfirmation.



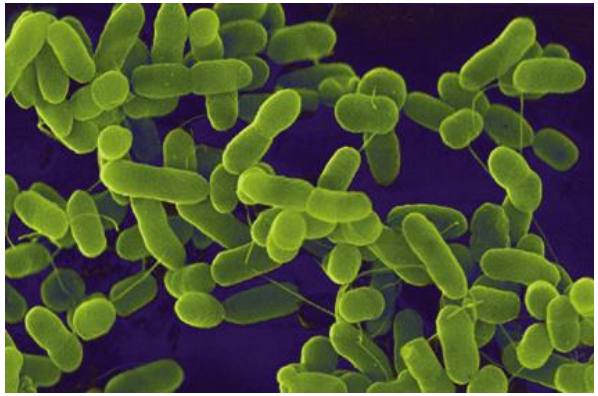
MacConkey agar

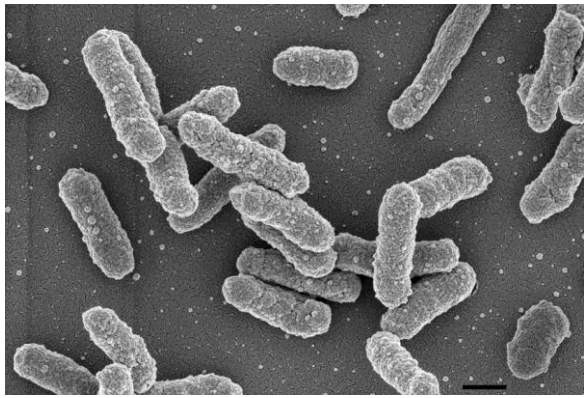
XLD agar

Treatment, prevention and control

หายเองได้

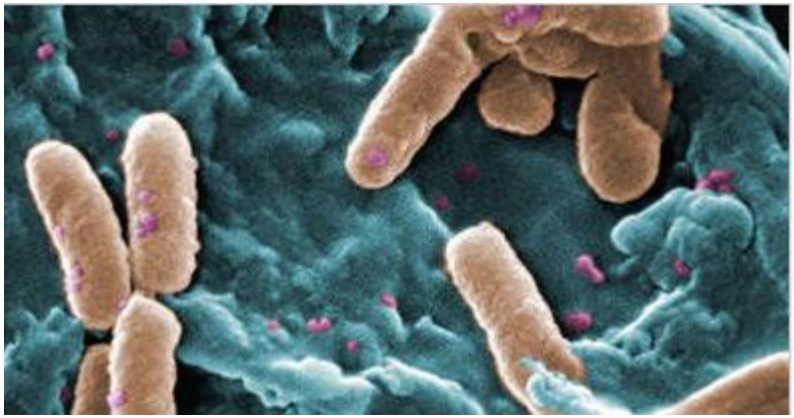
- Infection is generally self-limited.
- Antibiotic therapy shortens the course of symptomatic disease and fecal shedding.
- Treatment should be guided by in vitro susceptibility test.
- Ciprofloxacin, ceftriaxone and azithromycin have been used, depending on susceptibility testing.
- Appropriate infection control; prevent spread of organism, including hand washing and proper disposal of soiled linens



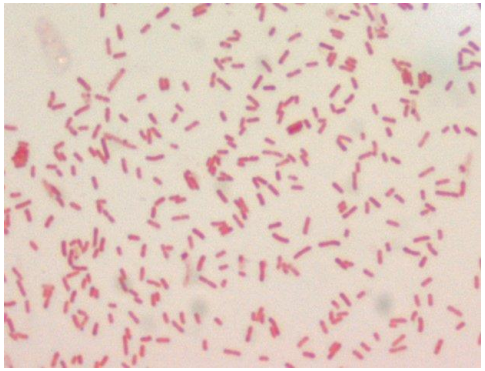


Yersinia spp.

(Family Yersiniaceae)



General characteristics

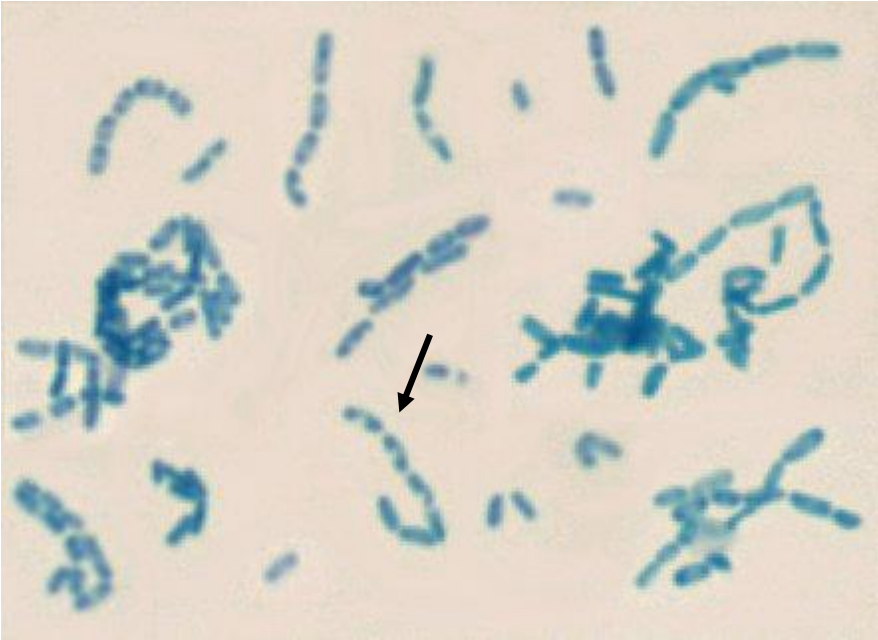


- Plump Gram-negative, facultative anaerobic rods
- ! Non-lactose-fermenter; oxidase negative !
- All possess O, some strains possess H.
- Motile at 22-29 ^{o}C by peritrichous flagella, except *Y. pestis* (non-motile)

  เหมือน Salmonella

  non-motile
- Optimal growth at 25-30 ^{o}C and some **can grow at 4 ^{o}C** !
- Bipolar staining when stained with Giemsa, Wright, methylene blue or Wayson (may be poor using Gram stain)
- Consists of 14 species, but only 3 species are considered human pathogens (***Y. pestis***, ***Y. enterocolitica*** and ***Y. pseudotuberculosis***)

General characteristics



Wayson stain

Wright stain

"Biopolar staining or safety pin appearance"

Pathogenesis and virulence factors

- ! *Y. pestis* is an etiologic agent of highly fatal systemic disease known as **plague**. ภาษีโรค โรคระบาดที่ทำให้เกิดการตายมาก
- *Y. enterocolitica* and *Y. pseudotuberculosis* are ! primarily enteric ! pathogens.
- A common characteristic of pathogenic *Yersinia* is its ability to ! **resist phagocytic killing** !
- On contact with phagocytes, the bacteria secrete Yop proteins that dephosphorylate several proteins required for phagocytosis.

bac จับกับ phagocytes $\xrightarrow[\text{Yop proteins}]{\text{Bac ปล่อย}}$ dephosphorylate proteins

ที่จำเป็นสำหรับ phagocytosis

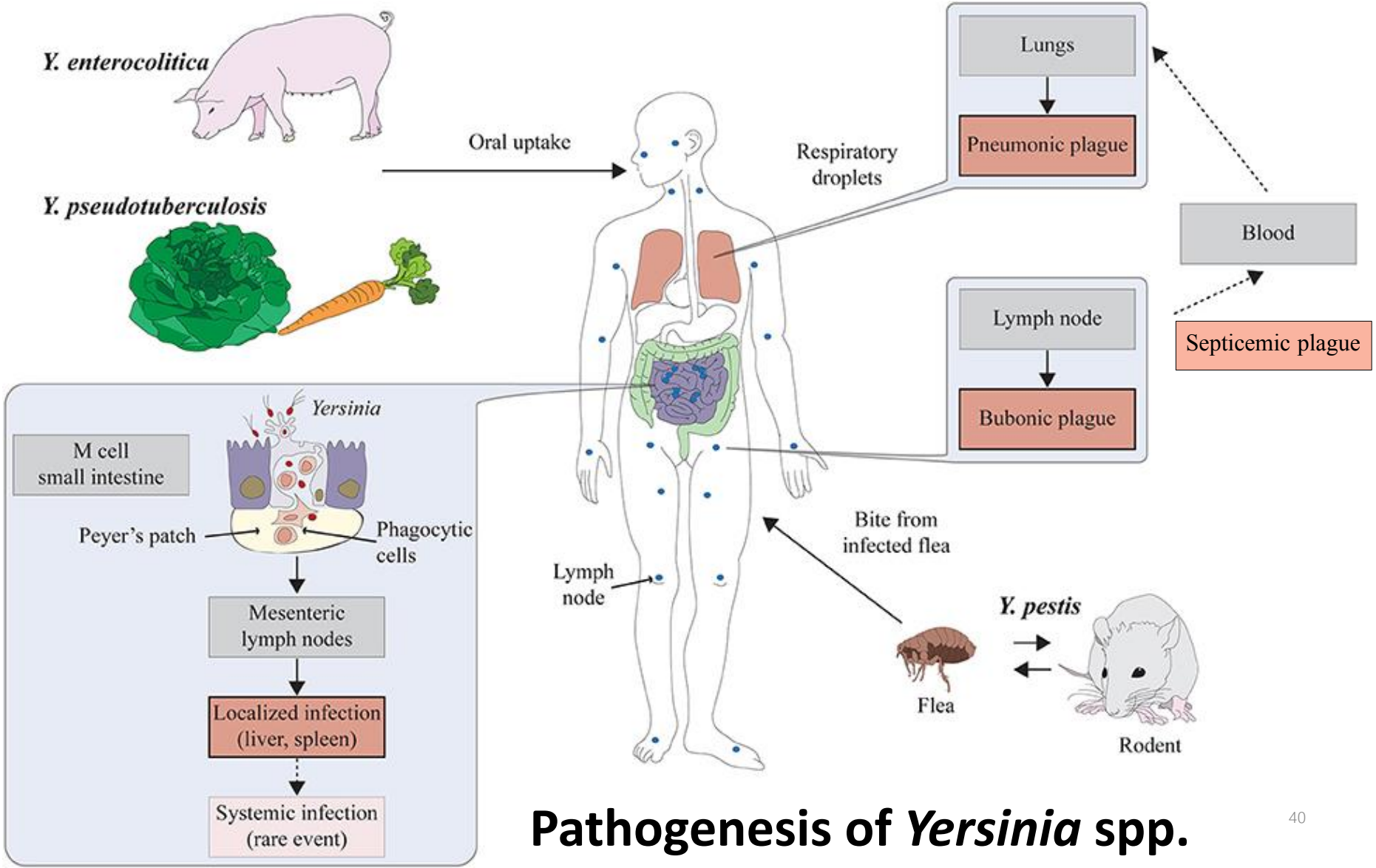
Pathogenesis and virulence factors

- Initiate apoptosis in macrophages

ยับยั้ง การสร้าง cytokines ลดการ ตอบสนอง inflammation

- Suppress cytokine production, which in turn diminishing inflammatory response to the infection

- *Y. pestis* has two plasmids that encode virulence genes:
  - Fraction 1 (F1) gene: **antiphagocytic protein capsule**
  - **Plasminogen activator protease** (*pla*) gene สลาย fribin ที่ลม cell
  - Other virulence factors: serum resistance and iron acquisition by siderophore-independent mechanism



Pathogenesis of *Yersinia* spp.

Clinical manifestations

- There are 3 major forms of plaque:
  - ① **Bubonic plague**
    - Incubation period; <7 days after infected flea bites
    - ขวบน้ำเหลือง
    - High fever and a painful bubo (inflammation swelling of lymph nodes) in the groin and axilla
    - แบคทีเรียอยู่ในกระแสเลือด
    - Bacteremia develops rapidly; if no treatment, 75% of patients die
  - ② **Pneumonic plague**
    - Incubation period; 2-3 days
    - Fever, malaise and pulmonary signs develop within 1 day

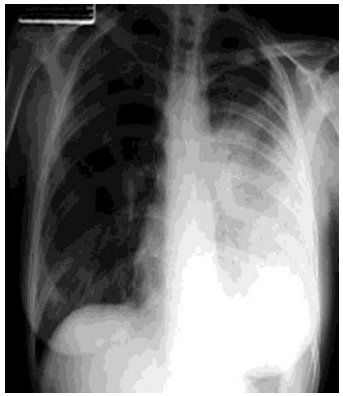
Clinical manifestations

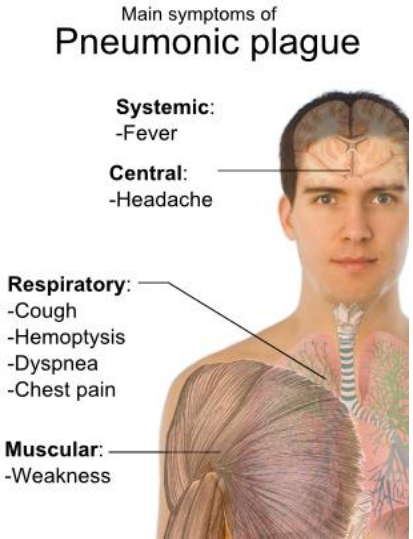
- The patients are highly infectious; person to person spread occurs by aerosols
- In untreated patients; mortality rate >90%

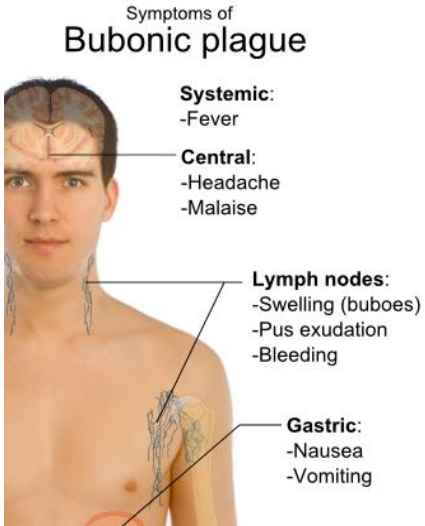
3

Septicemic plaque

- Patients develop fever, chills, extreme weakness, abdominal pain, shock and bleeding into the skin.







Clinical manifestations

- Other *Yersinia* spp. cause **gastroenteritis**.
  - Infective dose: 10^8-10^9 bacteria (*Y. enterocolitica*)
  - Incubation period 1-10 days
  - diarrhea, fever and abdominal pain
  - Other symptoms; septicemia, arthritis, intra-abdominal abscess, hepatitis and osteomyelitis
  - Enteric disease in children may manifest as enlarged mesenteric lymph nodes and **mimic acute appendicitis**.

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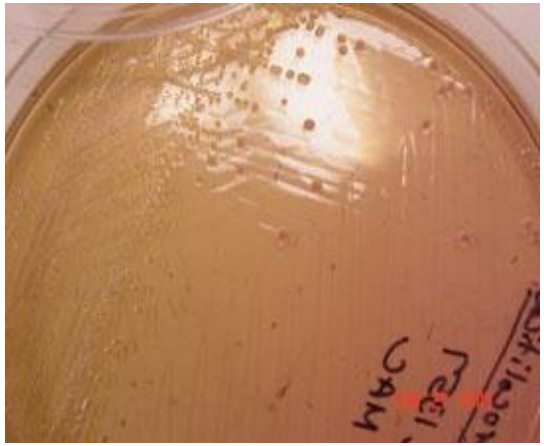
Epidemiology

- All *Yersinia* infections are **zoonotic** that humans are the accidental host.
- *Y. pestis* is spread by **flea** bites or direct contact with infected tissues or person to person by inhalation of infectious aerosols.
- Plague epidemics have occurred in Africa, Asia and South America.
- Other *Yersinia* infections are spread through exposure to contaminated food products.
- *Y. enterocolitica* infections have occurred worldwide, predominantly in Europe. The natural reservoirs are pig, rodents, livestock and rabbits.
- *Y. psuedotuberculosis* is a relatively uncommon cause of human disease. The natural reservoirs are rodents, wild animals and birds.

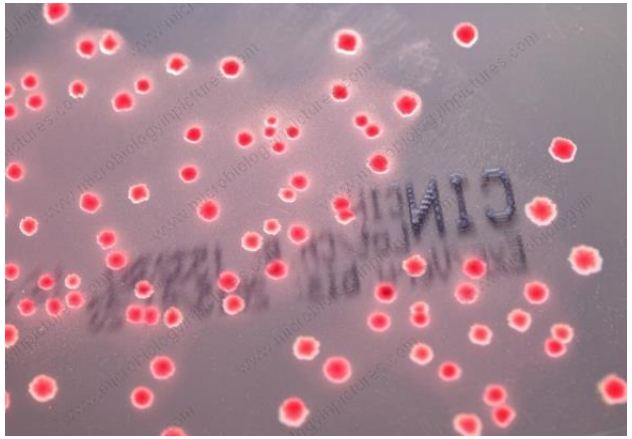
Laboratory Diagnosis

- Isolation from blood, lymph nodes aspirates, sputum or stool
- Organisms grow on most culture media. Prolonged storage at 4 ^{O}C can selectively enhance isolation (cold enrichment) in case of *Y. enterocolitica*.
- ✳ Cefsulodin-irgasan-novobiocin (CIN) agar is a selective medium for *Y. enterocolitis*. ❗
- Isolates were identified with biochemical tests.

Blood agar



MacConkey agar



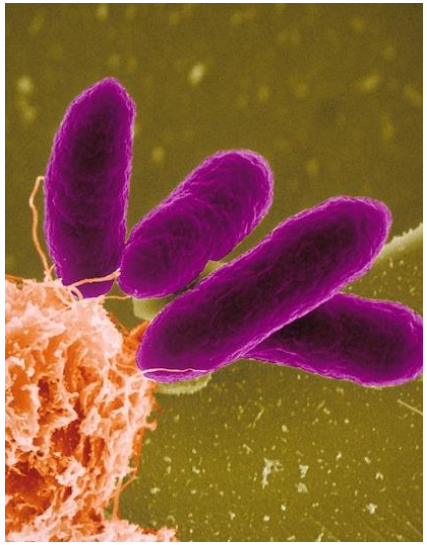
CIN agar

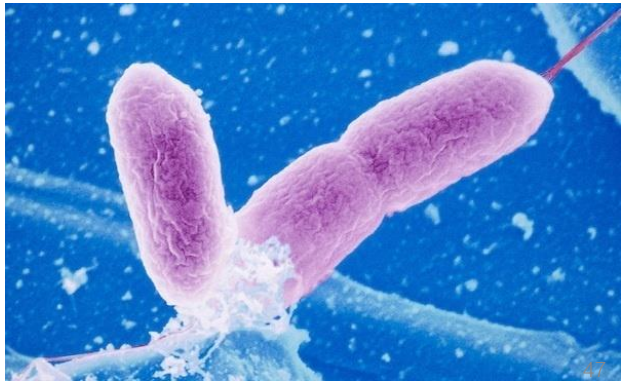
Treatment, prevention and control

- Streptomycin is the drug of choice for *Y. pestis* treatment, while tetracyclines, chloramphenicol or trimethoprim-sulfamethoxazole are as alternatives.
- Enteric infections with *Yersinia* spp. are usually self-limited.
  - *Y. pseudotuberculosis* is susceptible to ampicillin, cephalosporins, aminoglycosides and tetracyclines.
  - Aminoglycosides or trimethoprim sulfamethoxazole are drugs of choice for *Y. enterocolitica*.
- Plague is controlled by reduction of the rodent population and vaccination of individuals at risk.
- Other *Yersinia* infections are controlled by proper preparation of food products.

Plesiomonas shigelloides

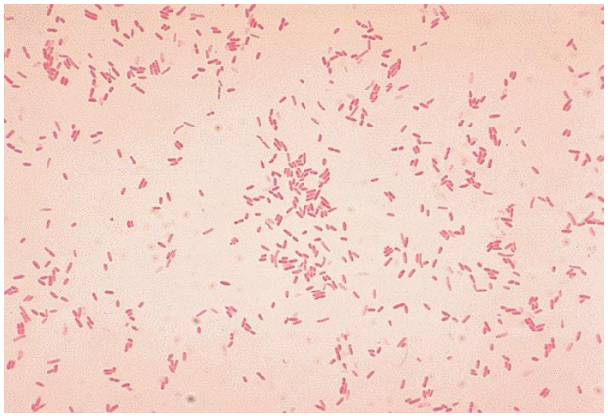
(Unclassified family)





General characteristics

- Gram-negative, facultative anaerobic rods
- Non lactose-fermenter; **oxidase positive**
- Possesses O and H antigens
- Motile by polar flagella
- Does not produce gas from glucose
- Ferment inositol
- LD, OD and AD are positive.
- Grow in 0% NaCl but does not grow in 6% NaCl
- Share biochemical and antigenic features with Shigellae



Classification

- The genus *Plesiomonas* was formerly in the family Vibrionaceae.
- Recent phylogenetic studies have shown that it is actually closely related to the members of the family Enterobacteriaceae.
- It is the only oxidase-positive member in the family Enterobacteriacae.
- ***P. shigelloides*** is still the only species in this genus.
- Plesiomonads often cross-agglutinate with Shigellae.

Pathogenesis and virulence factors

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- *P. shigelloides* is not considered to be part of normal human flora.
- *P. shigelloides* appear to possess a much lower virulence potential than *Shigella*, with a low symptomatic carriage rate among humans.
- Only a few possible virulent features:
  - Heat-stable toxin
  - Heat-labile toxin
  - Cholera-like toxin

Clinical manifestations

1

- Gastroenteritis
  - The infectious dose is presumed to be >10^6 CFU
  - Incubation period: 1-4 days
  - The more common **watery or secretory diarrhea** !
  - A subacute or chronic disease that lasts from 14 days to 3 months
  - A more invasive, dysenteric form that resembles colitis
  - 25- 40% of patients present with fever, vomiting or both
  - The most common clinical symptom is **abdominal pain**. !
  - Most cases are self-limiting.

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Clinical manifestations

- Extraintestinal infection
  - The risk groups are veterinarians, zookeepers, aquaculturists, fish handler and athletes.
  - Infections include bacteremia, meningitis, and wound infections.
  - More severe infections usually occur only in severely immunocompromised patients or neonates.
  - Biliary tract disease has been identified as possible risk factor for bacteremia with this organism.

Epidemiology

- *P. shigelloides* is found in both soil and aquatic environments (**only in fresh and estuarine water** of tropical and subtropical climates).
- They are widely distributed among both warm- and cold-blooded animals.
- They have emerged as a potential cause of enteric disease in humans and have also been isolated from a number of extraintestinal infections.
- Most of reported cases occur in adults.
- Association with the ingestion of **contaminated water, raw or undercooked seafood**, particularly clams, oysters or shrimp.

Laboratory diagnosis

- Isolation from stool, blood or CSF specimens
- *Plesiomonas* grows readily on most media routinely used, but it ! does not grow on TCBS agar. !
- Isolates were identified with biochemical tests.

Blood agar

MacConkey agar

Treatment, prevention and control

- Most cases are usually self-limiting, antibiotic therapy is indicated in patients with severe or chronic gastroenteritis.
- *P. shigelloides* is intrinsically resistant to penicillins due to the presence of a β-lactamase.
- Fluoroquinolones and trimethoprim-sulfamethoxazone are first-line agents for gastrointestinal illness.
- Extraintestinal infections, particularly among neonates, often require antibiotic therapy.
- Systemic infections have been successfully treated with fluoroquinolones, carbapenems or β-lactam/inhibitor combinations.

Laboratory diagnosis of Enterobacterales

- Specimen collection and transport
  - Transport media; Cary-Blair, Amies or Stuart media
- Isolation and identification
  - Direct microscopic examination
    - ! Stool samples; is not useful !
    - Other specimens from sterile sites; is very useful
  - Culture
    - Most grow at an optimal temperature of 35-37 ^{O}C
    - Certain species can grow at low temperatures (*Yersinia* and *Serratia*)
    - Organisms can grow on most media routinely used (BA, MAC, XLD).

Laboratory diagnosis of Enterobacterales

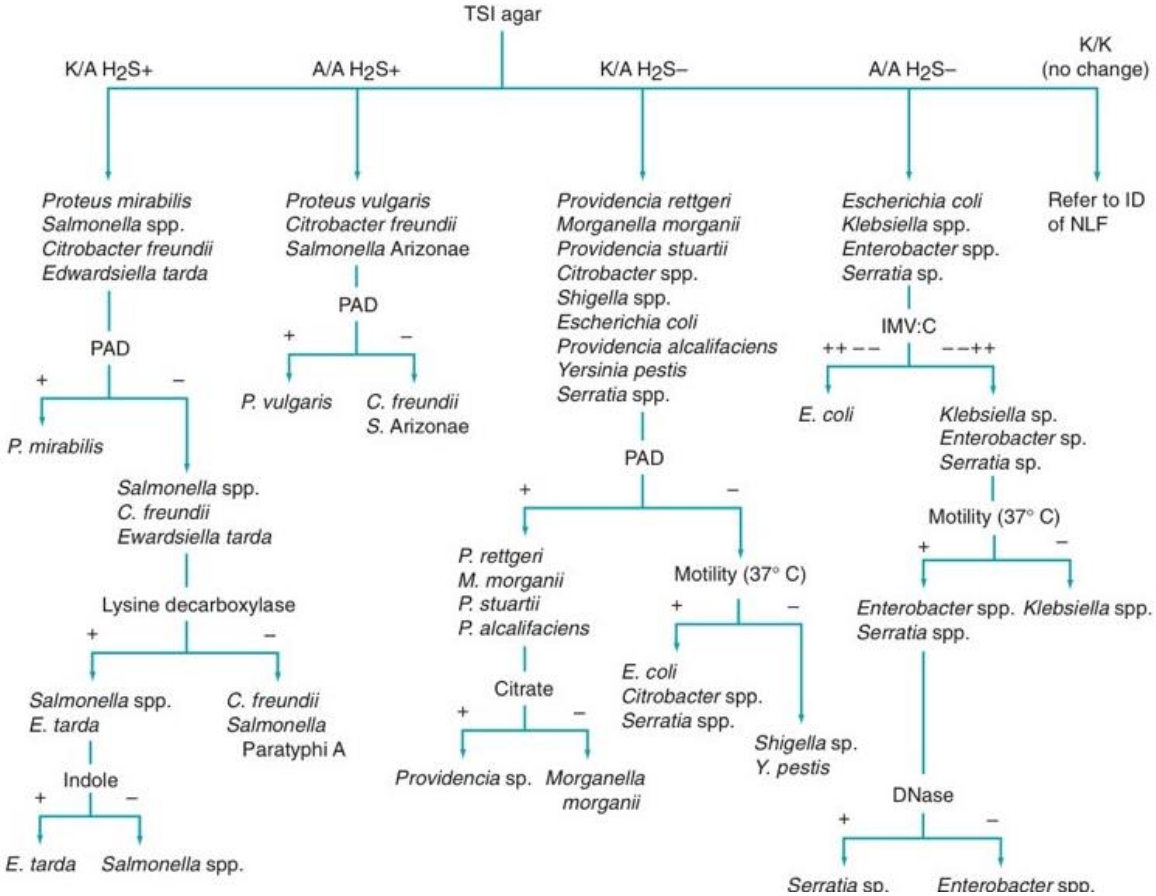
- Identification
  - Biochemical test (TSI, motile, indole, MR, VP, citrate, malonate, PD, urease, gas from glucose, mannitol, arabinose, LD, OD, DNase)
  - MALDI-TOF

- Serologic grouping
  - *Salmonella*; based on common O antigens (serogroup A-I), H antigen typing is usually performed in a reference laboratory
  - *Shigella*; based on O antigen (serogroup A-D)
  - Pathogenic *E. coli*: based on O and H antigens
  - *Plesiomonas*; based on O and H antigens

Serogrouping for *Salmonella* spp.

- Test with normal saline
  - No agglutination
- Test with *Salmonella* O Polyvalent (A-I) antisera
  - If positive, test with OMA and OMB polyvalent antisera, respectively
    - If positive with OMA, test with A, B, D, E and I antisera
    - If positive with OMB, test with C, F, G and H antisera
    - Report; *Salmonella* group A-I
  - If negative, test with Vi antisera
    - If positive, suspend in normal saline and heat at 100 ^{O}C for 15-60 min, then test again with O polyvalent antisera

Flow chart for the presumptive identification of Enterobacterales



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